



Faculty of Computers & Artificial Intelligence
CS213 - Object Oriented Programming

Dr. Mohamed Elramly

First Semester - Year 2025-2026

Assignment 3: TUI game application

Section: 17, 18

Team:

Name	ID	Stud-Gmail
Ammen Osama	20240745	20240745@stud.fci-cu.edu.e g
Samir Kahlawy	20240251	
Abdallah Youssef	20240331	
Mostafa Ahmed	20240588	

A3 - TA: Dalia Maher

Work breakdown:

Ammen: 4x4, pyramid, ultimate, menu + **Extras:** maze, snake, spaceship

Samir: SUS, numerical, 4 in a row, memory tic tac toe.

Mostafa: Misere, Diamond, infinity, documentation, readme.

Abdallah: Obstacles, 5x5, Word tic tac toe, Memory Tic Tac Toe.

The work was done fully through github with a lot of collaboration on maintaining clean branches and separation of concerns so we can avoid merge conflicts as much as possible.

“A PIE” concepts:

- Abstraction: for example, the menu abstracts away the functionality of the games and only deals with an interface namespace that contains functions that handle the game’s instantiation and state.
- Inheritance: the classes we created inherit from already-made classes provided by the doc. In all games, the UI and Board classes of each respective game is inherited from the provided abstract classes that the doctor made. In some cases we made our own classes (example: token class in 4x4 tic tac toe) to solve problems that are very specific to the game itself, without ever modifying the provided header file by the doctor.
- Polymorphism: A great example of polymorphism would be the `setup_players()` function in each of our games. It overrides the

setup_player() function that is provided by the doctor's header file to accommodate the needs of each game.

- Encapsulation: this happens in almost every game we have. Some nice examples would be the token class in the 4x4 tic tac toe game, which encapsulates the token positions and only lets you edit them through getters and setters.

Some Game screenshots:

```
test, enter your move:
Enter letter (A-Z): T
Enter position x and y (0-2): 0 2
Placed letter 'T' at position (0, 2)
```

```
      0   1   2
-----
0 | C | A | T |
-----
1 | E | . | . |
-----
2 | B | . | . |
-----
```

Obstacles and word game:

```
Valid word found in row 0: CAT

*** test wins by forming a valid 3-letter word! ***
Press any character to continue...
```

```
Please enter your move x and y (0 to 5): 0 4
```

```
      0       1       2       3       4       5
-----
0 |   X |   . |   0 |   . |   X |   . |
-----
1 |   . |   . |   . |   . |   . |   . |
-----
2 |   # |   . |   . |   . |   . |   . |
-----
3 |   . |   . |   . |   . |   . |   . |
-----
4 |   . |   . |   . |   # |   . |   . |
-----
5 |   . |   . |   . |   . |   . |   . |
-----
```

```
Please enter your move x and y (0 to 5):
```

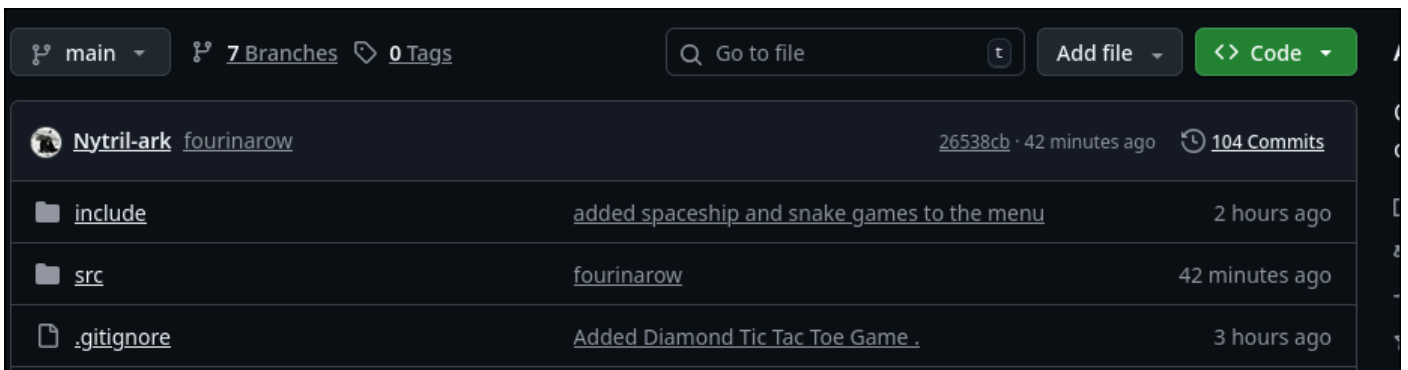
[illegible]

Github:

Please note:

As I am the owner of the repo I have to approve a lot of the pull requests, so my name might appear a lot more than usual in the commit history as each PR approval counts as a commit.

There were many branches each with their own commit history, and over 100 commits, so we cant include them all here. These are some of them:



The screenshot displays the GitHub repository interface for a project. At the top, the 'main' branch is selected, with 7 branches and 0 tags. A search bar labeled 'Go to file' is present. Below the repository name 'Nytril-ark', the commit history is shown. The first commit is by 'fourinarow' (26538cb, 42 minutes ago) with 104 commits. The commit history table lists the following changes:

File	Commit Message	Time
include	added spaceship and snake games to the menu	2 hours ago
src	fourinarow	42 minutes ago
.gitignore	Added Diamond Tic Tac Toe Game .	3 hours ago

main 7 Branches 0 Tags

Go to file

Add file

<> Code

About



Switch branches/tags

Find or create a branch...

Branches Tags

✓ main

default

Diamond_Game

abdallah

ammen

infinity_tic_tac_toe

misere

samir

View all branches

README.md

Word Tic Tac Toe.jpg

dic.txt

labyrinth.jpg

nyril-ark/ammen

6e51e1b · 1 hour ago

103 Commits

added spaceship and snake games to the menu

1 hour ago

test

1 hour ago

Added Diamond Tic Tac Toe Game .

3 hours ago

ammen: added extra games like the labyrinth/dijkstra ga...

17 hours ago

misere tic tac toe . remaining AI player and computer pla...

5 days ago

Organized project structure and added .gitignore

3 weeks ago

Initial commit

3 weeks ago

Rename Menu .jpg to Menu.jpg

1 hour ago

Rename Obsatcles Tic Tac Toe.jpg to Obsatcles Tic Tac T...

1 hour ago

Fix image tag for Menu screenshot in README

1 hour ago

Rename Word Tic Tac Toe.jpg to Word Tic Tac Toe.jpg

1 hour ago

WordTicTacToe(V1)

2 days ago

Add files via upload

1 hour ago

README

MIT license



Fun Box Collection

A comprehensive C++ game framework featuring 15+ strategic board games built on the BoardGame_Classes.h architecture

Game Dev assignment as part of the oop class in FCAI - CU 2025

[Readme](#)[MIT license](#)[Activity](#)[0 stars](#)[0 watching](#)[0 forks](#)

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 4

[Nytril-ark](#)[engmostafacs50](#) Mostafa Ahmed[Abdullah-Youssef22](#)[samirkahlawy](#) Samir Kahlawy

Languages

C++ 97.8% CMake 1.2% C 1.0%

Game 3 : 5*5 TicTacToe(V2)

 [Abdullah-Youssef22](#) committed 4 days ago

pressToFinish

 [Nytril-ark](#) authored 4 days ago

added random player . remaining AI player.

 [engmostafacs50](#) committed 4 days ago

Game 3 : 5*5 TicTacToe(V1)

 [Abdullah-Youssef22](#) committed 4 days ago

minor change

 [Nytril-ark](#) committed 4 days ago

 Commits on Nov 24, 2025

finished game 7: 4x4 tic tac toe

 [Nytril-ark](#) committed 5 days ago

misere tic tac toe . remaining AI player and computer player

 [engmostafacs50](#) committed 5 days ago

Update README.md

 [Nytril-ark](#) authored 5 days ago

 Commits on Nov 23, 2025

trying to solve this segfault

 [Nytril-ark](#) committed last week

token tracking and debugging funcs

 [Nytril-ark](#) committed last week

token system

 [Nytril-ark](#) committed last week

Commits on Nov 27, 2025	
samir: porting fix for sus constructor Nytril-ark committed 2 days ago	6a5107b
testing PR: animated megalovania character in the menu ... Nytril-ark authored 2 days ago	Verified a60624b
Merge branch 'main' into ammen Nytril-ark authored 2 days ago	Verified 68a50c8
added megalovania character to menu for fun Nytril-ark committed 2 days ago	2627ed2
Merge pull request #20 from Nytril-ark/abdallah ... Nytril-ark authored 2 days ago	Verified a70c976
WordTicTacToe(V1) Abdullah-Youssef22 committed 2 days ago	ca4fed8
WORD GMAE (DRAFT 1) Abdullah-Youssef22 committed 2 days ago	6061de1
5*5 ... Abdullah-Youssef22 committed 2 days ago	86ad260

Refactor SUS Board constructor and add setup_players in .h samirkahlawy authored 3 days ago	Verified 5823b03
Refactor SUS Board constructor and add player setup samirkahlawy authored 3 days ago	Verified a128aad
Merge PR from ammen: finished both 4x4 tic tac toe (game 7) and infinity tic tac toe (game 11) ... Nytril-ark authored 4 days ago	Verified 1b6644a

Commits on Nov 28, 2025	
My local work samirkahlawy committed yesterday	
Commits on Nov 27, 2025	
Merge pull request #25 from Nytril-ark/ammen ... Nytril-ark authored 2 days ago	
samir: porting fix for sus constructor	

Video link:

<https://youtu.be/O4I-3vBPRDM>

Code review:

While some of my colleagues were reviewing the code I wrote for the menu, they discovered a bug in the menu that makes it clear the screen before the winner of the game is announced. As such, me and my friends each implemented a fix for that bug in our respective game folders, that works with the nature of each game we're making.

Moreover, we made sure to review any changes that happened to any files in the github PR page before they are merged into the main branch. In some cases where those changes were going to damage the project, we end the code review with a disapproval until the person at hand fixes that.

Gladly, there weren't many such conflicts as we achieved nearly a complete separation of concerns, and the only cases where we had to review each other's code were in the single conflict point which is the menu clearing bug. We tested each other's games, and in some cases where the local library used in the game wasn't found, we would notify each other to include it so that the game is completely cross platform.